

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279830

Kind Code

A1

Publication Date

September 04, 2025

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OPTICAL COMMUNICATION SYSTEM FOR PASSIVE OPTICAL NETWORKS AND HIGH DATA RATE COMMUNICATION

Abstract

A bi-directional optical communication system employing a minimum number of single-mode high repetition rate pulsed optical signal sources to achieve cost efficiency while maintaining high data rates. The bi-directional optical communication system includes a first optical data processing unit and a second optical data processing unit. The first optical data processing unit modulates a pulsed optical source using a differential quadrature phase shift keying (DQPSK) modulation and two-level pulse amplitude (PAM-2) modulation and then demodulates it to achieve a pulse amplitude modulated signal. The second optical data processing unit reuses the same optical carrier by passing it through a regenerative wavelength converter to generate three pulsed optical carriers at different wavelengths and employs an On-off keying (OOK) modulation scheme. These carriers are employed to send uplink data at a same rate of as the downlink. As a result, large data is transmitted from one data center to another data center through a downlink and uplink free space optical link network.

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Appl. No.: 19/208654

Filed: May 15, 2025

Related U.S. Application Data

Publication Classification

Int. Cl.: **H04B10/112** (20130101); **H04B10/079** (20130101); **H04B10/40** (20130101); **H04B10/508** (20130101); **H04B10/516** (20130101); **H04B10/524** (20130101); **H04B10/54** (20130101); **H04B10/556** (20130101); **H04B10/572** (20130101); **H04B10/61** (20130101)

U.S. Cl.:

CPC **H04B10/1125** (20130101); **H04B10/0799** (20130101); **H04B10/40** (20130101); **H04B10/508** (20130101); **H04B10/5161** (20130101); **H04B10/524** (20130101); **H04B10/54** (20130101); **H04B10/5561** (20130101); **H04B10/572** (20130101); **H04B10/612** (20130101); **H04B10/613** (20130101); **H04B10/6163** (20130101); **H04B10/6164** (20130101);

Background/Summary

STATEMENT REGARDING PRIOR DISCLOSURE BY THE INVENTORS

[0001] Aspects of the invention are described in an article “A Novel 60 Gbps Bidirectional Free Space Optical Link Based on a Single Laser Source” published in Arabian Journal for Science and Engineering on Jun. 12, 2022, which is incorporated herein by reference in its entirety.

BACKGROUND

Technical Field

[0002] The present disclosure is directed to a bi-directional optical communication system. In particular, the optical communication system disclosed herein provide a bi-directional free space optical link-based communication in a passive optical network and a high data rate and cost-efficient transmission of data.

Description of Related Art

[0003] The “background” description provided herein is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description which may not otherwise qualify as prior art at the time of filing, are neither expressly or impliedly admitted as prior art against the present invention.

[0004] Data centers (DCs) have become essential as a vital information technology (IT) infrastructure in the current technological era. DCs are facilities that include data processing devices such as servers, networking devices, storage systems, computing infrastructure, and the like, networked for storing and distributing data and/or building, running, and delivering applications and services. The DCs facilitate technologies such as cloud computing, Internet technologies and services including online gaming, search engines, e-commerce, social networking, video streaming, and the like.

[0005] Currently, there may be a multitude of servers in a DC that are connected to each other through intra-data center interconnects that usually span over a distance of less than, for example, 300 m [See: Simpanen, E.; Gustavsson, J. S.; Larsson, A.; Karlsson, M.; Bickham, S. R.: 1060 nm single-mode VCSEL and single-mode fiber links for long-reach optical interconnects. J. Lightwave Technol. 37(13), 2963-2969 (2019), incorporated herein by reference in its entirety]. Transceivers

for such short-reach intra-data center interconnects are designed in an 850 nm window employing pulse amplitude modulation (PAM)-4 modulated data, optical multimode (OM) $\frac{3}{4}$ multimode fiber (MMF)-based channel, and Gallium Arsenide (GaAs)-based vertical-cavity surface-emitting lasers (VCSELs) as light sources [See: Simpanen, E.; Gustavsson, J. S.; Larsson, A.; Karlsson, M.; Bickham, S. R.: 1060 nm single-mode VCSEL and single-mode fiber links for long-reach optical interconnects. *J. Lightwave Technol.* 37(13), 2963-2969 (2019), incorporated herein by reference in its entirety].

[0006] For long-reach inter-data center interconnects that span over distances larger than 2 km, single-mode fibers (SMFs) are used in the 1310 nm or 1550 nm wavelengths with Silicon photonics or InP transceivers [See: Denoyer, G.; Cole, C.; Santipo, A.; Russo, R.; Robinson, C.; Li, L.; Zhou, Y.: Hybrid silicon photonic circuits and transceiver for 50 Gb/s NRZ transmission over single-mode fiber. *J. Lightwave Technol.* 33(6), 1247-1254 (2015), incorporated herein by reference in its entirety]. Optical fibers have several limitations when deployed to connect DCs that are separated by, for example, several kilometers. Some of the limitations include high costs of installation and maintenance, considerable deployment time, immobility of DCs, and high outage probability due to fiber-related faults.

[0007] Conventional DCs are mostly located in buildings that normally take long periods sometimes even several years to assemble. Ideally, the DCs should be easy to move and install to meet flexible business requirements. As a result, a modular data center (MDC) was introduced. An MDC is placed in, for example, a shipping-based container that is usually 20 ft to 40 ft in length and includes a multitude of servers, storage devices, network devices, uninterrupted power supply, cooling systems etc. The MDCs can be adjusted to run independently or to become part of a larger DC by connecting with other MDCs to meet different business requirements. MDCs offer a host of benefits including mobility, modularity, easy transportation, rapid deployment, low power usage, reduced capital expenditure (CAPEX) and operational expenditure (OPEX).

[0008] Owing to the aforementioned limitations of optical fibers and its usage with MDCs, free space optics (FSO) based inter-data center interconnects for MDCs have been considered as a viable alternative to realize mobility, easy transportation, rapid deployment, energy efficiency and cost efficiency of DCs [See: Chen, T.; Gao, X.; Chen, G.: The features, hardware, and architectures of data center networks: a survey. *J. Parallel Distrib. Comput.* 96, 45-74 (2016), incorporated herein by reference in its entirety]. Free-space optical communication is an optical communication technology that uses light beams (mostly in line of sight) propagating in free space to wirelessly transmit and receive voice, video, and data information.

[0009] FSO communication has recently gained significant research interest for deployment as an access network, both for front-haul and back-haul scenarios [See: Chen, T.; Gao, X.; Chen, G.: The features, hardware, and architectures of data center networks: a survey. *J. Parallel Distrib. Comput.* 96, 45-74 (2016), incorporated herein by reference in its entirety]. The various advantages of FSO communication include large bandwidth, no license required for spectrum utilization, reduced consumption of power, smart hardware, secure information transmission, and protection from electromagnetic interference (EMI). The FSO communication can be implemented at a fraction of the deployment cost of optical fiber-based communication systems [See: Mirza, J.; Ghafoor, S.; Hussain, A.: All-optical generation and transmission of multiple ultrawideband signals over free space optical link. *Opt. Eng.* 58(5), 056103 (2019), incorporated herein by reference in its entirety].

[0010] Optical access networks, particularly passive optical networks, radio over fiber, and fiber-to-the-home systems, have large setting-up and maintenance costs besides the long-time span needed for the aforementioned activities [See: Mirza, J.; Ghafoor, S.; Hussain, A.: All-optical generation and transmission of multiple ultrawideband signals over free space optical link. *Opt. Eng.* 58(5), 056103 (2019) and Bhatnagar, M. R.; Ghassemlooy, Z.: Performance analysis of Gamma-Gamma Fading FSO MIMO links with pointing errors. *J. Lightwave Technol.* 34, 2158-2169 (2016), incorporated herein by reference in its entirety]. The cost and time factor escalates

significantly, for example, when the terrain is congested and obstructed by buildings, roads, housing, and shopping malls, which is the case in metropolitan cities. The tasks of digging the ground, laying down the fiber, installing the network components as well as their maintenance are cumbersome, expensive, time-consuming, and cause disturbances to the public. The FSO communication significantly eliminates all these tasks, therefore lowering the cost of implementation, adding simplicity, and in reducing time for implementation.

[0011] However, the performance of the FSO link at times suffers from atmospheric attenuation and random variation of signal intensity and phase due to atmospheric turbulence, commonly known as intensity scintillation [See: Mirza, J.; Ghafoor, S.; Hussain, A.: All-optical generation and transmission of multiple ultrawideband signals over free space optical link. *Opt. Eng.* 58(5), 056103 (2019), incorporated herein by reference in its entirety]. Intensity scintillation is similar to multi-path fading in wireless communication and originates from random fluctuations of the refractive index of the atmosphere caused by variations in air temperature and pressure [See: Gappmair, W.; Flohberger, M.: Error performance of coded FSO links in turbulent atmosphere modeled by Gamma-Gamma distributions. *IEEE Trans. Wirel. Commun.* 8(5), 2209-2213 (2009), incorporated herein by reference in its entirety].

[0012] Various FSO statistical channel models are available for the estimation of turbulence, but the most frequently used statistical channel models are the Log-normal model, K model, Negative exponential model, Gamma-Gamma model and Log-normal Rician model [See: Andrews, L. C.; Phillips, R. L.: *Laser beam propagation through random media*, vol. 152. SPIE Press, Bellingham, WA (2005), incorporated herein by reference in its entirety]. The Gamma-Gamma channel model is a generalized FSO channel model that is widely used for modeling of atmospheric turbulence and attenuation in all turbulence regimes, such as from weak to strong.

[0013] For the past few years, FSO communication systems delivering high data rate signals to remote users employing Log-normal and Gamma-Gamma channel models have been widely researched. An FSO link based on On-Off keying (OOK) modulated signal to investigate the improvement in performance due to coherent detection and dynamic decision threshold in the presence of atmospheric turbulence [See: Wang, Z.; Zhong, W.-D.; Yu, C.: Performance improvement of OOK free-space optical communication systems by coherent detection and dynamic decision threshold in atmospheric turbulence conditions. *IEEE Photonics Technol. Lett.* 24(22), 2035-2037 (2012), incorporated herein by reference in its entirety].

[0014] Hybrid intensity modulation and direct detection FSO communication system operating in visible wavelength regime is disclosed in [See: Yang, F.; Gao, J.; Liu, S.: Novel visible light communication approach based on hybrid OOK and ACO-OFDM. *IEEE Photonics Technol. Lett.* 28(14), 1585-1588 (2016), incorporated herein by reference in its entirety]. The study integrates asymmetrically clipped optical orthogonal frequency division multiplexing and OOK modulation formats to enhance spectral efficiency. The performance of FSO communication links with and without turbulence using hybrid modulation formats such as quadrature amplitude modulation and multi-pulse position modulation technique is investigated in [See: Khallaf, H. S.; Shalaby, H. M.; Garrido-Balsells, J. M.; Sampei, S.: Performance analysis of a hybrid QAM-MPPM technique over turbulence-free and Gamma-Gamma free-space optical channels. *IEEE/OSA J. Opt. Commun. Netw.* 9(2), 161-171 (2017), incorporated herein by reference in its entirety].

[0015] Efficient detection of OOK modulated signals at the receiver to maximize the bandwidth efficiency in FSO communication systems requires an instantaneous channel fading coefficient which is implemented through a blind data detection method [See: Dabiri, M. T.; Sadough, S. M. S.: Generalized blind detection of OOK modulation for free-space optical communication. *IEEE Commun. Lett.* 21(10), 2170-2173 (2017), incorporated herein by reference in its entirety]. The performance of an FSO link is analyzed for differential phase shift keying (DPSK) modulation scheme with and without combining Manchester coding [See: Padhy, J. B.; Patnaik, B.: Design and analysis of multiplexed FSO system with DPSK and Manchester coding. In: 2017 3rd International

Conference on Applied and Theoretical Computing and Communication Technology (iCATccT), pp. 1-6. IEEE (2017), incorporated herein by reference in its entirety].

[0016] Integration of FSO with a passive optical network was disclosed and experimentally demonstrated by transmitting 10 Gbps OOK data [See: Yeh, C. H.; Chow, C. W.; Gu, C. S.; Guo, B. S.; Cheng, Y. J.; Chen, J. H.: Performance analysis of free space optical communication traffic integrated with passive optical network. *Electron. Lett.* 54(21), 1228-1229 (2018).

<https://doi.org/10.1049/el.2018.5559>, incorporated herein by reference in its entirety]. An all-optical relay-assisted FSO communication system based on amplify and forward relay is experimentally demonstrated, where the performance is analyzed for OOK and DPSK modulation formats [See: Huang, X.; Xie, X.; Song, J.; Duan, T.; Hu, H.; Xu, X.; Su, Y.: Performance comparison of all-optical amplify-and-forward relaying fso communication systems with OOK and DPSK modulations. *IEEE Photonics J.* 10(4), 1-11 (2018), incorporated herein by reference in its entirety].

[0017] A low-cost and high data rate scheme for photonic generation and transmission of multiple ultra-wideband signals to four different radio access units by using a centralized frequency comb source has been described [See: Mirza, J.; Ghafoor, S.; Hussain, A.: All-optical generation and transmission of multiple ultrawideband signals over free space optical link. *Opt. Eng.* 58(5), 056103 (2019), incorporated herein by reference in its entirety]. A cost-efficient integrated FSO architecture combining multiple services for passive optical networks was experimentally demonstrated by using a single laser source located at the central office [See: Yeh, C.-H.; Lin, W.-P.; Luo, C.-M.; Xie, Y.-R.; Chang, Y.-J.; Chow, C.-W.: Utilizing single lightwave for delivering baseband/FSO/MMW traffics simultaneously in PON architecture. *IEEE Access* 7, 138927-138931 (2019), incorporated herein by reference in its entirety].

[0018] The conventional communication systems that utilize free space optical links comprise complex processing techniques that result in complex architectures. To achieve a required data rate for communication between data centers, a complex technique utilizing a high component count is implemented in the conventional communication architectures. Generally, for generation of each FSO link a separate source is used, resulting in multiple FSO link sources in a communication architecture. To account for the turbulence that occurs over optical communication networks is another challenge faced by modelling such optical communication systems. This causes problems in the implementation of the communication models when reduced to practice.

[0019] Accordingly, a free-space optical link-based communication system, as disclosed herein, is utilized for bi-directional data communication in a passive optical network that employs a combination of modulation schemes to transmit high data rate signals over a turbulent free-space optical channel. The bi-directional optical communication system, as disclosed herein, provides high data rates and cost-efficient communication while using simple modulation schemes that do not require complex detection techniques and a single laser source with wavelength regenerator to reduce component count.

SUMMARY

[0020] In an embodiment, a bi-directional optical communication system is described. The bi-directional optical communication system includes a first optical data processing unit and a second optical data processing unit. The first optical data processing unit includes a first transmitter. The first transmitter includes a first channel, a second channel, a differential quadrature phase shift keying (DQPSK) encoder, a Gaussian pulse generator, an optical power splitter, at least two Mach-Zehnder modulators (MZM), a phase shifter, an optical combiner, a third channel, an Electro-absorption Modulator (EAM) and an optical amplifier. The first channel is configured to provide a first electrical signal at a first data rate. The second channel is configured to provide a second electrical signal at the first data rate. The differential quadrature phase shift keying (DQPSK) encoder is configured to encode the first electrical signal and the second electrical signal to provide at least two differentially encoded electrical signals at the first data rate. The Gaussian pulse

generator is configured to generate an optical pulsed signal centered at a wavelength with a pulse duration. The optical power splitter is configured to split the optical pulsed signal into at least two optical pulsed signals. The at least two MZM are configured to phase modulate the at least two optical pulsed signals using the at least two differentially encoded electrical signals to provide a first phase modulated optical pulsed signal and a second phase modulated optical pulsed signal. The phase shifter is configured to phase shift the first phase modulated optical pulsed signal to generate a phase shifted phase modulated optical pulsed signal. The optical combiner is configured to combine the second phase modulated optical pulsed signal and the phase shifted phase modulated optical pulsed signal to generate a processed optical pulsed signal at a second data rate. In examples, the second data rate is two times the first data rate. The third channel is configured to provide a third electrical signal at the first data rate. The EAM is configured to amplitude modulate the processed optical pulsed signal using the third electrical signal to provide a phase-amplitude modulated optical pulsed signal at a third data rate. In some examples, the third data rate is three times the first data rate. The optical amplifier is configured to amplify the pulse-amplitude modulated optical pulsed signal and generate a transmitter optical pulsed signal for transmission. The transmitter optical pulsed signal is transmitted as a first free space optical link through a first transmitter telescope. The second optical data processing unit includes a first receiver, a first optical splitter, a second optical splitter, a photodetector, a differential quadrature phase shift keying (DQPSK) demodulator, at least three low-pass filters, and at least three bit-error rate analyzers. The first receiver includes the first receiver telescope that is configured to receive an optical pulsed signal. The receiver optical pulsed signal is an attenuated version of the transmitter optical pulsed signal. The first optical splitter is configured split the optical pulsed signal into a receiver optical pulsed signal and a reuse optical pulsed signal. The second optical splitter is configured to split the receiver optical pulsed signal into a first optical pulsed signal, a second optical pulsed signal and a third optical pulsed signal. The photodetector is configured to convert the first optical pulsed signal into a first electrical signal with two amplitude levels corresponding to the third channel in the first transmitter.

[0021] The DQPSK demodulator is configured to demodulate the second optical pulsed signal and the third optical pulsed signal into a first amplitude-modulated electrical signal and a second amplitude-modulated electrical signal. The at least three low-pass filters are configured to remove harmonics from the first electrical signal, the first amplitude-modulated electrical signal and the second amplitude-modulated electrical signal to provide three filtered electrical signals. The at least three bit-error rate analyzers are configured to estimate a bit error rate of each of the three filtered electrical signals.

[0022] The second optical data processing unit further includes a wavelength converter configured to utilize the reuse optical pulsed signal for further transmission. The second optical data processing unit includes a wavelength converter, and a second transmitter. The wavelength converter includes a first optical amplifier, an optical bandpass filter, and a highly nonlinear fiber (HNLF). The first optical amplifier is configured to receive the reuse optical pulsed signal and further configured to suppress amplitude variations of the reuse optical pulsed signal. The optical bandpass filter is tuned at the wavelength configured to remove out-of-band noise and generate a filtered optical pulsed signal. The noise removed herein, is in the form of an amplified spontaneous emission (ASE) noise.

[0023] The HNLF is configured to self-phase modulation-induced spectral broadening of the filtered optical pulsed signal to generate a converted optical pulsed signal. The first optical amplifier is configured to receive the reuse optical pulsed signal and further configured to suppress amplitude variations of the reuse optical pulsed signal. The optical bandpass filter is tuned at the wavelength configured to remove out-of-band noise and generate a filtered optical pulsed signal. The HNLF is configured to self-phase modulation-induced spectral broadening of the filtered optical pulsed signal to generate a converted optical pulsed signal.

[0024] The second transmitter includes an optical splitter, at least three optical bandpass filters configured, at least three channels, at least three Mach-Zehnder modulators (MZM), an optical combiner, and a second optical amplifier. The optical splitter is configured to split the converted optical pulsed signal into three optical pulsed signals. The at least three optical bandpass filters are configured to filter the three optical pulsed signal into three filtered optical pulsed signals. Each of the at least three optical bandpass filters are at an off set from the wavelength. The at least three channels wherein each channel provides an electrical signal at a first data rate. The at least three Mach-Zehnder modulators MZM is configured to On-Off keying (OOK) modulate the three filtered optical pulsed signals using the three electrical signals to provide three modulated optical pulsed signals. The optical combiner is configured to combine the three modulated optical pulsed signals into a modulated optical pulsed signal. The second optical amplifier is configured to amplify the modulated optical pulsed signal and generate a second transmitter optical pulsed signal. The second transmitter optical pulsed signal is transmitted as a second free space optical link through a second transmitter telescope. As described above, the second free space optical link is a modified version of the first free space optical link, achieved through the use of the wavelength converter.

[0025] The first optical data processing unit further includes a second receiver. The second receiver includes a second receiver telescope, an optical splitter, at least three optical bandpass filters, at least three photodetectors, at least three low-pass filters, and at least bit error rate analyzers. The second receiver telescope is configured to receive an optical pulsed signal. The optical splitter is configured to split the received optical pulsed signal into three optical pulsed signals. The at least three optical bandpass filters centered at the off set from the wavelength is configured to filter the three optical pulsed signals to provide three filtered optical pulsed signals. The at least three photodetectors are configured to convert the three filtered optical pulsed signals into three electrical signals. The at least three low-pass filters are configured to remove harmonics of the three electrical signals. The at least bit error rate analyzers are configured to estimate a bit error rate of the three electrical signals.

[0026] In some examples, the wavelength is 1552 nm.

[0027] The electro-absorption modulator (EAM) utilizes a non-return-to-zero (NRZ) pulsed signal at the first data rate to implement pulse amplitude modulation. The phase-amplitude modulated optical pulsed signal is encoded as a differential quadrature phase shift keying (DQPSK) and two-level pulse amplitude modulation (PAM-2).

[0028] The differential quadrature phase shift keying (DQPSK) demodulator includes a plurality of delay lines, a plurality of phase shifters, and a group of four detectors. The plurality of phase shifters are configured to compare a bit of a receive optical pulsed signal to a previous bit of the received optical pulsed signal. The group of four detectors are configured to employ balanced detection of the received optical pulsed signal.

[0029] In examples, the three filtered electrical signals in the first receiver correspond to a data of the first channel, the second channel and the third channel in the first transmitter.

[0030] The wavelength converter is a self-phase modulation (SPM)-based wavelength converter is configured to reuse a received optical signal for further transmission of electrical signals.

[0031] In examples, the off set of the wavelength are three wavelength values including 1550.6 nm, 1552.5 nm, and 1553.3 nm.

[0032] The three electrical signals at the second receiver correspond to the electrical signals of the three channels at the second transmitter.

[0033] The optical bandpass filter of the wavelength converter is configured to remove noise in the form of an amplified spontaneous emission (ASE) noise.

[0034] The first optical data processing unit and the second optical data processing unit are each at least one selected from a data center and a modular data center.

[0035] The first free space optical link is a point-to-point free space optical link.

[0036] The first free space optical link is at a data rate of three times the first data rate.

[0037] The first free space optical link is at least one from a unidirectional free space link and a bidirectional free space link.

[0038] The second free space optical link is a modified version of the first free space optical link.

[0039] The foregoing general description of the illustrative embodiments and the following detailed description thereof are merely exemplary aspects of the teachings of this disclosure, and are not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0040] A more complete appreciation of this disclosure and many of the attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings, wherein:

[0041] FIG. 1A shows an architecture where two modular data centers (MDCs) are deployed in a line-of-sight communication for establishing a point-to-point bi-directional FSO link.

[0042] FIG. 1B shows a multi-tier hierarchical tree-based architecture of the MDCs.

[0043] FIG. 2A—illustrates a first part of an exemplary block diagram for a bi-directional optical communication system, according to certain embodiments.

[0044] FIG. 2B illustrates a second part of an exemplary block diagram for a bi-directional optical communication system, according to certain embodiments.

[0045] FIG. 3 is a plot showing attenuation versus input voltage of Electro-absorption Modulator (EAM), according to certain embodiments.

[0046] FIG. 4 is a plot illustrating average power of output of optical bandpass filters at different wavelengths versus average power at an input of the highly nonlinear fiber (HNLF), according to certain embodiments.

[0047] FIG. 5A shows an eye diagram of a received optical signal at an MDC after transmission through a free space optics (FSO) link, according to certain embodiments.

[0048] FIG. 5B shows an eye diagram of a demodulated in-phase signal, according to certain embodiments.

[0049] FIG. 5C shows an eye diagram of a demodulated quadrature signal, according to certain embodiments.

[0050] FIG. 5D shows an eye diagram of the received optical signal observed after wavelength converter centered at 1550.6 nm, according to certain embodiments.

[0051] FIG. 5E shows an eye diagram of the received optical signal after the wavelength converter centered at 1552.5 nm, according to certain embodiments.

[0052] FIG. 5F shows an eye diagram of the received optical signal after the wavelength converter centered at 1553.5 nm, according to certain embodiments.

[0053] FIG. 5G shows an eye diagram of a direct detected uplink signal for a first channel, according to certain embodiments.

[0054] FIG. 5H shows an eye diagram of a direct detected uplink signal for a second channel, according to certain embodiments.

[0055] FIG. 5I shows an eye diagram of a direct detected uplink signal for a third channel, according to certain embodiments.

[0056] FIG. 6 shows bit error rate (BER) versus received optical power plots for the downlink and uplink signals under medium turbulence conditions.

DETAILED DESCRIPTION

[0057] In the drawings, like reference numerals designate identical or corresponding parts throughout the several views. Further, as used herein, the words “a,” “an” and the like generally

carry a meaning of “one or more,” unless stated otherwise.

[0058] Furthermore, the terms “approximately,” “approximate,” “about,” and similar terms generally refer to ranges that include the identified value within a margin of 20%, 10%, or preferably 5%, and any values therebetween.

[0059] The disclosure describes a high data rate, bidirectional optical communication system for data centers (DCs) and/or modular data centers (MDCs) based on a single laser source that may be placed at any the DC or MDC. FIG. 1A-FIG. 1B show an exemplary scenario of FSO-based MDC architecture **150**. FIG. 1A shows an architecture comprising two MDCs (MDC **102** and MDC **104**) deployed in a line-of-sight communication for establishing a point-to-point bi-directional FSO link **106**. FSO-based inter-data center interconnects are used for bidirectional transmission of data between the MDCs (**102**, **104**), and the MDCs (**102**, **104**) are linked with the internet **112** through their respective optical gateways (OGs **108**, **110**). The optical gateway is a piece of hardware that provides an interface between MDCs and external network.

[0060] FIG. 1B shows a multi-tier hierarchical tree-based architecture **150** of the MDCs (**102**, **104**). The multi-tier hierarchical tree-based architecture **150** herein is described for the MDC **102**, which is equally applicable to the MDC **104**. The multi-tier hierarchical tree-based architecture **150** includes one or more servers **116.sub.1-X** (where X is a positive integer) that are communicatively coupled by top of the rack switches **114.sub.1-Y** (where Y is a positive integer). The top of the rack switches **114.sub.1-Y** may refer to switches that communicatively couple the one or more servers **116.sub.1-X**, appliances and/or switches in a same rack or adjacent or nearby racks. In some scenarios, the top of rack switches **114.sub.1-Y** are communicatively coupled by leaf switches **124.sub.1-124.sub.N**. The leaf switches **124.sub.1-124.sub.M** are devices used in the MDC **102** to aggregate traffic from the server nodes and connect to the core of the MDC **102** that includes spine switches **126.sub.1-126.sub.M**. The spine switches **126.sub.1-126.sub.M** include switches that perform traffic routing, and function as core of the MDC **102**. The MDCs (**102**, **104**) are communicatively coupled through border leaf switches. For example, border leaf switch **124.sub.N** is communicatively coupled with border leaf switch **134.sub.N** of the MDC **104** through bi-directional FSO link for communication. Also, border leaf switch **124.sub.N-1** couples with border leaf switch **134.sub.N-1** of the MDC **104** through bi-directional fiber link **132** (for example, single mode fiber) for communication. The communication between the MDC **102** and the MDC **104** may be at the border leaf switches (**124.sub.N** and **134.sub.N** and/or **124.sub.N-1** and **134.sub.N-1**) either through the bi-directional fiber link **132** and/or through FSO-based inter-data center interconnects. The deployment of FSO for inter-data center interconnects results in reduced expenditure and low power consumption while maintaining high data rates.

[0061] Referring to FIG. 2A-FIG. 2B that illustrates an exemplary block diagram for a bi-directional optical communication system **200** using free space optical link, according to one or more embodiments. The bi-directional optical communication system **200** may also be implemented as a unidirectional optical communication system in an embodiment of the present disclosure.

[0062] The bi-directional optical communication system **200** includes a first optical data processing unit **201** as shown in FIG. 2A, and a second optical data processing unit **241**, as shown in FIG. 2B. In an example, the first optical data processing unit **201** and the second optical data processing unit **241** may correspond to MDCs (for example, the MDC **102**, the MDC **104**). In another example, the first optical data processing unit **201** and the second optical data processing unit **241** may correspond to data centers. In some examples, the first optical data processing unit **201** and the second optical data processing unit **241** each at least one selected from a data center and the MDCs.

[0063] The first optical data processing unit **201** is illustrated in FIG. 2A. The second optical data processing unit **241** is illustrated in FIG. 2B. As shown in FIG. 2A, the first optical data processing unit **201** includes a first transmitter **202**. The first transmitter **202** includes a first channel **204**, a second channel **206**, a differential quadrature phase shift keying (DQPSK) encoder **208**, a Gaussian

pulse generator **210**, an optical power splitter **212**, at least two Mach-Zehnder modulators (MZM) **214**, **216**, a phase shifter **218**, an optical combiner **220**, a third channel **222**, an Electro-absorption Modulator (EAM) **224** and an optical amplifier **226**. It is also to be appreciated that while bi-directional optical communication system **200** is utilized for illustrative purposes of the example novel embodiments of the present disclosure, it is to be understood that other alternative commercial and custom configurations, as briefly stated above, having various other components can also be incorporated when using the techniques of the present application is capable of being operated within the design parameters of the invention.

[0064] In an embodiment of the present disclosure, the first channel **204**, the second channel **206** and the third channel **222** are configured to provide a first electrical signal, the second electrical signal, and the third electrical signal, respectively, at a first data rate. In the context of the present disclosure, the first channel **204**, the second channel **206** and the third channel **222**, may refer to any means such as, wireless medium, wires, such as, for example, coaxial cables, twisted pair cables, and the like, for communicating electrical signals. The DQPSK encoder **208** may be a unit configured to implement DQPSK modulation technique on the signals. The DQPSK is a modulation technique in which a carrier may assume one of four phases, each change of phase, or symbol, representing 2 bits. The bit combinations may include 00, 01, 11 and 10. As a result of DQPSK, data stream may carry 2 bits at a time. In the current context, DQPSK encoder **208** is configured to encode the first electrical signal and the second electrical signal to provide at least two differentially encoded electrical signals at the first data rate. Each of the at least two differentially encoded electrical signals is further split into two electrical signals.

[0065] The Gaussian pulse generator **210** is configured to generate an optical pulsed signal centered at a wavelength with a pulse duration. In some examples, the wavelength is 1552 nm. In the embodiments disclosed herein, the Gaussian pulse generator **210** as the term suggests, is a pulse generator that creates a sequence of optical pulses modulated by an input digital signal. The optical power splitter **212** is configured to split the optical pulsed signal into at least two optical pulsed signals. In examples, the optical power splitter **212** may refer to an integrated waveguide optical power distribution unit configured to split an incident light beam into two or more light beams, and vice versa, containing multiple input and output ends.

[0066] The at least two MZM **214**, **216** are configured to phase modulate the at least two optical pulsed signals using the at least two differentially encoded electrical signals to provide a first phase modulated optical pulsed signal and a second phase modulated optical pulsed signal. As shown in FIG. 2A, the encoded electrical signals that are further split into two electrical signals are provided to corresponding MZMs. For example, the signals corresponding to the encoded electrical signals (from source channel **204**) is provided to MZM **214**. Similarly, the signals corresponding to the encoded electrical signals (from source channel **206**) is provided to MZM **216**. In an example, the at least two MZM **214**, **216** may be made from a material with electro-optic effect (for example, LiNbO₃, GaAs, InP). Each of the MZM **214** and the MZM **216** includes two 'arms' that splits an input signal. When electric fields are applied to the arms, the MZM **214**, and the MZM **216** change optical path lengths resulting in phase modulation. The arms are combined at the end, which combines the signals with different phases (for example, phase modulation) and converts the phase modulation into intensity modulation.

[0067] The phase shifter **218** is configured to phase shift the first phase modulated optical pulsed signal to generate a phase shifted phase modulated optical pulsed signal. In examples, the phase shifter **218** is a unit that provides phase shift by introducing a defined amount of delay to the input signal. The optical combiner **220** is configured to combine the second phase modulated optical pulsed signal and the phase shifted phase modulated optical pulsed signal to generate a processed optical pulsed signal at a second data rate. In examples, the second data rate is two times the first data rate. The optical combiner **220** is a passive unit in which signals from several output fibers is received and then distributed to one or more input fibers or photoreceivers. The EAM **224** is

configured to amplitude modulate the processed optical pulsed signal using the third electrical signal to provide a phase-amplitude modulated optical pulsed signal at a third data rate. In some examples, the third data rate is three times the first data rate. In examples, the EAM **224** utilizes a non-return-to-zero (NRZ) pulsed signal at the first data rate to implement pulse amplitude modulation. The phase-amplitude modulated optical pulsed signal may be encoded as a differential quadrature phase shift keying (DQPSK) and two-level pulse amplitude modulation (PAM-2). In examples, the EAM **224** is a device that may be used for modulating the intensity of a laser beam via an electric voltage.

[0068] The optical amplifier **226** is configured to amplify the pulse-amplitude modulated optical pulsed signal and generate a transmitter optical pulsed signal for transmission. In examples, the optical amplifier **226** is a unit that amplifies light without converting an optical signal to an electrical signal. A first transmitter telescope **228** transmits the transmitter optical pulsed signal as a first free space optical (FSO) link **230**. In examples of the embodiments disclosed herein, the first FSO link **230** is a point-to-point free space optical link. In some examples of the embodiments disclosed herein, the first FSO link **230** may have a data rate of three times the first data rate. In examples of the embodiments disclosed herein, the first FSO link **230** is at least one from a unidirectional free space link and a bidirectional free space link.

[0069] Referring to FIG. 2B, the second optical data processing unit **241** includes a first receiver **242**, a first optical splitter **244**, a second optical splitter **246**, a photodetector **254**, a differential quadrature phase shift keying (DQPSK) demodulator **255**, at least three low-pass filters **258.sub.1-3**, and at least three bit-error rate analyzers **260.sub.1-3**. The first receiver **242** is configured to receive the optical pulsed signal. The first receiver **242** includes the first receiver telescope **243**. The first receiver telescope **243** is configured to receive an optical pulsed signal. The receiver optical pulsed signal is an attenuated version of the transmitter optical pulsed signal. The attenuation is due to turbulence or disturbances caused in the atmosphere.

[0070] The first optical splitter **244** may split the received pulsed optical signal. The first optical splitter **244** splits the optical pulsed signal into a receiver optical pulsed signal **245** and a reuse optical pulsed signal **247**. The second optical splitter **246** is configured to split the receiver optical pulsed signal into a first optical pulsed signal **248**, a second optical pulsed signal **250** and a third optical pulsed signal **252**. The first optical pulsed signal **248** is transmitted to the photodetector **254** that converts the first optical pulsed signal **248** into a first electrical signal **257** with two amplitude levels corresponding to the third channel **222** in the first transmitter **202**. The second optical pulsed signal **250** and the third optical pulsed signal **252** are input to the DQPSK demodulator **255**.

[0071] The DQPSK demodulator **255** is configured to demodulate the second optical pulsed signal **250** and the third optical pulsed signal **252** into a first amplitude-modulated electrical signal and a second amplitude-modulated electrical signal, respectively. The DQPSK demodulator **255** includes optical splitters **256.sub.1-256.sub.2**, a plurality of delay lines **280.sub.1-280.sub.2**, a plurality of phase shifters **294.sub.1-294.sub.2**, optical combiners **283.sub.1-283.sub.2** and a group of four detectors **284.sub.1-284.sub.4**. The optical splitter **256.sub.1** splits the second optical pulsed signal into two signals. Similarly, the optical splitter **256.sub.2** splits the third optical pulsed signal into two signals. Each of the signals are input to corresponding delay lines **280.sub.1-280.sub.2** and phase shifters **294.sub.1-294.sub.2**. The plurality of delay lines **280.sub.1-280.sub.2** may add variable delays to the corresponding signals. For example, the delay line **280.sub.1** is configured to introduce a delay to the second optical pulsed signal. Similarly, the delay line **280.sub.2** is configured to introduce a delay to the third optical pulsed signal. In examples, the delay introduced by the delay lines **280.sub.1** is different from the delay line **280.sub.2**. The plurality of phase shifters **294.sub.1-294.sub.2** are configured to compare a bit of a received optical pulsed signal to a previous bit of the received optical pulsed signal. For example, the phase shifters **294.sub.1** is configured to phase shift the second optical pulsed signal. Similarly, the phase shifters **294.sub.2** is configured to phase shift the third optical pulsed signal. In an example, the phase shift may be a

quarter wavelength. The optical combiner **283.sub.1** combines the delayed second optical pulsed signal and the phase shifted second optical pulsed signal. Similarly, the optical combiner **283.sub.2** combines the delayed third optical pulsed signal and the phase shifted third optical pulsed signal. Each of the combined second optical pulsed signal and the combined third optical pulsed signal are further split into two signals and input to corresponding group of four detectors **284.sub.1-284.sub.4**. The group of four detectors **284.sub.1-284.sub.4** are configured to employ balanced detection of the received optical pulsed signals. For example, detectors **284.sub.1-284.sub.2** process the split signals from the combined second optical pulsed signal and combine to generate the first amplitude-modulated electrical signal **292.sub.1**. The detectors **284.sub.1-284.sub.2** process the split signals from the combined third optical pulsed signal and combine to generate the second amplitude-modulated electrical signal **292.sub.2**.

[0072] The at least three low-pass filters, including the first low-pass filter **258.sub.1**, the second low-pass filter **258.sub.2**, and the third low-pass filter **258.sub.3** are configured to remove harmonics from the first electrical signal **257**, the first amplitude-modulated electrical signal **292.sub.1** and the second amplitude-modulated electrical signal **292.sub.2**, respectively to provide three filtered electrical signals. The at least three bit-error rate (BER) analyzers including BER analyzer **260.sub.1**, BER analyzer **260.sub.2** and BER analyzer **260.sub.4** are configured to estimate a bit-error rate of each of the three filtered electrical signals. In aspects, the three filtered electrical signals in the first receiver correspond to data of the first channel **204**, the second channel **206** and the third channel **222** in the first transmitter **202**.

[0073] The second optical data processing unit **241** in FIG. 2B further includes the wavelength converter **261**, and a second transmitter **268**. The wavelength converter **261** is configured to utilize the reuse optical pulsed signal for further transmission. In examples, the wavelength converter **261** is a self-phase modulation (SPM)-based wavelength converter configured to reuse a received optical signal for further transmission of electrical signals. The wavelength converter **261** includes a first optical amplifier **262**, an optical bandpass filter **264**, and a highly nonlinear fiber (HNLF) **266**. The first optical amplifier **262** is configured to receive the reuse optical pulsed signal **247** and further configured to suppress amplitude variations of the reuse optical pulsed signal **247**. The optical bandpass filter **264** is tuned at the wavelength configured to remove out-of-band noise and generate a filtered optical pulsed signal **265**. In examples, the optical bandpass filter **264** of the wavelength converter **261** is configured to remove noise in the form of an amplified spontaneous emission (ASE) noise. The HNLF **266** is configured to self-phase modulation-induced spectral broadening of the filtered optical pulsed signal to generate a converted optical pulsed signal **267**.

[0074] In FIG. 2B, the second transmitter **268** includes an optical splitter **269**, at least three optical bandpass filters **270.sub.1-270.sub.3**, at least three channels **271.sub.1-271.sub.3**, at least three MZM **272.sub.1-272.sub.3**, an optical combiner **274**, and a second optical amplifier **276**. The optical splitter **269** is configured to split the converted optical pulsed signal into three optical pulsed signals. The at least three optical bandpass filters **270.sub.1-270.sub.3** are configured to filter the three optical pulsed signal into three filtered optical pulsed signals. Each of the at least three optical bandpass filters **270.sub.1-270.sub.3** are at an off-set from the wavelength. For example, for a signal received with a wavelength 1552 nm, the off set of the wavelength are three wavelength values including 1550.6 nm, 1552.5 nm, and 1553.3 nm. Each of at least three channels **271.sub.1-271.sub.3** provides an electrical signal at a first data rate. At least three MZM **272.sub.1-272.sub.3** are configured to use modulation scheme, for example, such as, On-Off keying (OOK) modulation to modulate the three filtered optical pulsed signals using the three electrical signals to provide three modulated optical pulsed signals. The optical combiner **274** is configured to combine the three modulated optical pulsed signals into a modulated optical pulsed signal. The second optical amplifier **276** is configured to amplify the modulated optical pulsed signal and generate a second transmitter optical pulsed signal **277**. The second transmitter optical pulsed signal **277** is transmitted as a second FSO link **279** through a second transmitter telescope **278**. In examples, the

second free space optical link **279** is a modified version of the first free space optical link.

[0075] The first optical data processing unit **201** in FIG. 2A further includes a second receiver **290**. The second receiver **290** includes a second receiver telescope **280**, an optical splitter **281**, at least three optical bandpass filters **282.sub.1-282.sub.3**, at least three photodetectors **284.sub.1-284.sub.3**, at least three low-pass filters **286.sub.1-286.sub.3**, and at least bit error rate analyzers **288.sub.1-288.sub.3**. The second receiver telescope **280** is configured to receive an optical pulsed signal. The optical splitter **281** is configured to split the received optical pulsed signal into three optical pulsed signals. The at least three optical bandpass filters **282.sub.1-282.sub.3** centered at the off-set from the wavelength are configured to filter the three optical pulsed signals to provide three filtered optical pulsed signals.

[0076] The at least three photodetectors **284.sub.1-284.sub.3** are configured to convert the three filtered optical pulsed signals into three electrical signals. In aspects, the three electrical signals may correspond to the electrical signals of the three channels at the second transmitter **268**. The at least three low-pass filters **286.sub.1-286.sub.3** are configured to remove harmonics of the three electrical signals. The at least BER analyzers **288.sub.1-288.sub.3** are configured to estimate bit error rates of the three electrical signals.

[0077] In the embodiments disclosed herein, the performance of the disclosed bi-directional optical communication system **200** was assessed by performing a simulation as described herein. The simulation environment used in the embodiments herein is a commercial tool. For the simulation, the parameters used are as described in Table 1 below.

TABLE-US-00001 TABLE 1 Parameters for simulation. Parameter details Value Bit rate per channel 20 Gbps Number of bits transmitted 1024 Repetition rate of Gaussian pulse 20 GHz generator Center wavelength of Gaussian pulse 1552 nm generator Pulse width of the Gaussian pulse 10 ps generator Insertion loss of DD-MZMs 3 dB RF switching voltage of DD-MZM 4 V Noise figure of optical amplifiers 4 dB Aperture diameter of TX/RX 5/20 cm telescope Beam divergence 2 mrad FSO attenuation 3 dB/km Responsivity of PINs 0.9 A/W Bandwidth of optical bandpass filters 0.25 nm at receiver Cut-off frequency of electrical low- 16 GHz pass filters at receiver Length of HNLF 1 km Nonlinear parameter of HNLF 18 W.sup.-1 km.sup.-1 Dispersion of HNLF -1.7 ps/nm/km Dispersion slope of HNLF ps/nm.sup.2/km

[0078] The simulation disclosed herein is described in conjunction with FIG. 2A-FIG. 2B for ease of understanding of flow of signals. At the first transmitter **202**, the first channel **204** and the second channel **206** each having a first data rate of for example, 20 Gbps were provided to the DQPSK encoder **208**. The DQPSK encoder **208** provided two differentially encoded electrical pulsed signals (bit streams) at a data rate of 20 Gbps. The encoded electrical pulsed signals were used to phase modulate a Gaussian optical pulsed signal (generated by the Gaussian pulse generator **210**) through the MZMs **214, 216** (for example, dual drive (DD)-MZMs). The output of the Gaussian pulse generator **210** having a repetition rate of 20 GHz centered at a wavelength of 1552 nm and having a pulse duration of around 10 ps were split into two optical pulsed signals using the optical power splitter **212**. The two optical pulsed signals were applied to optical inputs of the MZMs **214, 216**. The MZMs **214, 216** phase modulated the at least two optical pulsed signals to provide a first phase modulated optical pulsed signal and a second phase modulated optical pulsed signal based on the first electrical signal and the second electrical signal. The second phase modulated optical pulsed signal was phase shifted by 90° (through the phase shifter **218**) and combined with the first phase modulated optical pulsed signal by the optical combiner **220** at the second data rate. In examples, the second data rate was two times the first data rate. In other words, the combined signal is a pulsed DQPSK signal having a symbol rate of 20 symbols per second and a bit rate of 40 Gbps. The pulsed DQPSK signal was input to the EAM **224** along with a third signal at the first data rate (a rate of 20 Gbps).

[0079] In some example scenarios, attenuation induced over the optical signal by the EAM **224** decreases as the third signal (for example, applied voltage) is increased along a positive axis.

Attenuation versus input voltage plot **302** of the EAM **224** is as shown in FIG. 3, where attenuation of the optical signal is shown to be decreasing due as applied voltage (V) is increasing. This property of the EAM **224** has been used to vary the amplitudes of the optical pulses of the input DQPSK signal by applying the NRZ pulsed signal to the electrical input of the EAM **224**. The EAM **224** amplitude modulates the processed optical pulsed signal using the NRZ pulsed signal (also known as the third electrical signal) to provide a phase-amplitude modulated optical pulsed signal at a third data rate. In examples, the NRZ signal having a data rate of 20 Gbps represents the third channel **222** (also known to as downlink channel), and results in the application of two-level PAM (1-PAM) to the DQPSK modulated signal. As a result, the output optical signal of the EAM **224** has a combined data rate of 60 Gbps encoded as DQPSK and PAM-2 using a single pulsed optical source. The output pulsed optical signal was amplified through the optical amplifier **226** having a noise figure of 4 dB and transmitted via the first FSO link **230** through the first transmitter telescope **228** having 5 cm of aperture diameter.

[0080] The pulsed optical beam whose divergence is 0.5 mrad was received by the first receiver telescope **243** after traversing a distance of 500 m over the first FSO link **230**. The attenuation of the first FSO link **230** was kept at 3 dB/km, and the first FSO link **230** induced medium turbulence over the optical signal which may represent a scenario of data center environments. In the embodiments disclosed herein, the first FSO link **230** were modeled using the Gamma-Gamma channel model (provided below). The Gamma-Gamma channel model are efficient in accounting for the turbulences over free space optical communication links. The pulsed optical signal at the output of the first receiver telescope **243** was divided into two signals by the first optical splitter **244** to the receiver optical pulsed signal **245** and the reuse optical pulsed signal **247**. The receiver optical pulsed signal **245** was communicated to the DQPSK demodulator, which split the receiver optical pulsed signal **245** into three signals (that is the first optical pulsed signal **248**, the second optical pulsed signal **250** and the third optical pulsed signal **252**) through the second optical splitter **246**. The first optical pulsed signal **248** was processed by the photodetector **254** to perform direct detection of intensity modulation applied by the third channel **222** at the first transmitter **202**. In example, the photodetector **254** had a responsivity of 0.9 A/W and generated an electrical signal at the output having two amplitude levels representing the ones and zeros of the third channel **222**, respectively. The electric signal generated by the photodetector **254** was passed through the first low-pass filter **258.sub.1** having a cut-off frequency of 16 GHz to remove unwanted spectral components such as the harmonics generated due to the nonlinearity of the photodetector. The filtered electrical signal was communicated to the BER analyzer **260.sub.1** which estimated the error rate. The second optical pulsed signal **250** and a third optical pulsed signal **252** of the second optical splitter **246** were communicated to the DQPSK demodulator **255**. The demodulation was performed by employing balanced detection using four photodetectors **284.sub.1-284.sub.4**, each having a responsivity of 0.9 A/W. The outputs of the photodetectors **284.sub.1-284.sub.4** (also referred to as balanced detectors) were composed of amplitude-modulated electrical signals (the first amplitude-modulated electrical signal and the second amplitude-modulated electrical signal) that were passed through the first low-pass filter **258.sub.1**, the second low-pass filter **258.sub.2**, and the third low-pass filter **258.sub.3** to remove unwanted spectral components. The first amplitude-modulated electrical signal **292.sub.1** and the second amplitude-modulated electrical signal **292.sub.2** represent the data of the first channel **204** and the second channel **206**. The data were given to the corresponding BER analyzer **260.sub.1**, **260.sub.2** for error estimation.

[0081] After performing demodulation to extract the data of the three channels transmitted from MDC-1, the reuse optical pulsed signal may be reused for the transmission of data from MDC **104** toward MDC **102**. In an example, a SPM-based wavelength converter **261** was employed to reuse the received optical signal. Apart from performing wavelength conversion, the wavelength converter **261** suppressed the amplitude variations induced over the optical pulsed signal. As shown in FIG. 2B, the reuse optical pulsed signal was given as input to the first optical amplifier **262**. The

first optical amplifier **262** had a high gain of 32 dB and a noise figure of 4 dB. The optical bandpass filter (OBPF) **264** tuned at 1552 nm and having a bandwidth of 0.8 nm was placed after the first optical amplifier **262** to remove out-of-band noise in the form of ASE noise, which was added by the first optical amplifier **262** in the FSO link.

[0082] The reuse optical pulsed signal **247** at the output of the optical bandpass filter (OBPF) **264** was passed through the HNLF **266**, resulting in significant SPM-induced spectral broadening since the input signal has power. The length of the HNLF **266** was 1 km, and its attenuation was 2 dB/km. Furthermore, the non-linear parameter of HNLF was $18 \text{ W} \cdot \text{sup.} - 1 \text{ km} \cdot \text{sup.} - 1$, dispersion was -1.7 ps/nm/km , and dispersion slope was $0.023 \text{ ps/nm} \cdot \text{sup.} 2/\text{km}$. The broadened spectrum at the output of the HNLF **266** was split into three optical pulsed signals using the optical splitter **269**. The three optical pulsed signals from the optical splitter **269** were given to three different OBPFs **270.sub.1-270.sub.3** having wavelengths of 1550.6 nm, 1552.5 nm, and 1553.5 nm, respectively.

[0083] Filtering the broadened spectrum at wavelengths that are off-set from the center wavelength of the input signal results in a step-like transfer function, as shown in FIG. **4**. Such a transfer function may be employed to reduce intensity fluctuations over the optical pulses when the power of the input signal is properly adjusted. FIG. **4** shows the power at the output of the OBPFs **270.sub.1-270.sub.3** (i.e., **402**, **404** and **406**, representing off-set wavelengths of 1552.5 nm, 1550.6 nm, and 1553.5 nm, respectively) versus the power of the optical signal at the input of the OBPFs **270.sub.1-270.sub.3**. It is to be noted that the plots shown in FIG. **4** indicates that for input signal power variation between 20 and 25 dB, the variation in the power at the output of the OBPFs **270.sub.1-270.sub.3** is very small. Therefore, choosing a suitable value of input signal power such that it lies on an operating point may cause a significant reduction in the intensity variations of the pulsed optical signal at the output. This property of the SPM-based wavelength converter **261** was unitized to suppress the intensity variations generated over the PAM-2 signal received at MDC **104**. This allowed generation of three pulsed optical signals that are centered at different wavelengths and have reduced intensity fluctuations.

[0084] As shown in FIG. **2A-FIG. 2B**, the three optical signals were intensity modulated by three different uplink channels **271.sub.1-271.sub.3** each having a data rate of 20 Gbps. The intensity modulation was conducted by employing three single-drive MZMs **272.sub.1-272.sub.3** whose electrical inputs were connected to the 20 Gbps NRZ signals of the uplink channels. The OOK signals at the output of the MZMs **272.sub.1-272.sub.3** were combined using the optical combiner **274**, amplified using the second optical amplifier **276**, and transmitted over the second FSO channel **279** through the second transmitter telescope **278** that is similar to the first transmitter telescope **228** at the MDC **102**. The uplink FSO channel **230** has the same length and parameters as the downlink FSO channel. The signal at the output of the second receiver telescope **280** were split into three signals using the optical splitter **281**. The OBPFs **282.sub.1-282.sub.3** centered at wavelengths of 1550.6 nm, 1552.5 nm, and 1553.5 nm were used to filter out each of the OOK signals transmitted by MDC **104**. The OOK signal at the output of each optical filter were detected by using photodetectors **284.sub.1-284.sub.3** that had a responsivity of 0.9 A/W. The photodetector **284.sub.1-284.sub.3** outputs were passed through the low-pass filters **286.sub.1-286.sub.3** to remove harmonics and were given to the BER analyzers **288.sub.1-288.sub.3** for estimating the error rate.

[0085] The FSO link **230** modeled using the Gamma-Gamma channel model is described herein. The optical signals transmitted over FSO links may be impaired by various detrimental effects of free space where atmospheric turbulence, and attenuation are major contributors. Atmospheric turbulence exists due to arbitrary fluctuations in temperature and pressure of the air in small pockets along the propagation path of the signal. Turbulence may result in intensity variations of the received optical signal, an effect generally known as intensity scintillation that is a major source of impairment in the FSO link. Various channel models have been discussed to accurately express the intensity scintillation of the optical signal. A wide range of turbulence conditions can be

modeled using the Gamma-Gamma channel model. The atmospheric effects on the normalized intensity of light I may be characterized as small-scale α or large-scale β intensity fluctuations. The characterization may be based on the magnitude of intensity variations that follow a Gamma distribution whose probability density function PDF(I) may be expressed as:

$$[00001] \text{ PDF}(I) = \frac{2(\frac{\alpha}{\beta})^{\frac{\alpha}{\beta}}}{(\frac{\alpha}{\beta})^{\frac{\alpha}{\beta}} (\frac{\alpha}{\beta})^{\frac{\alpha}{\beta}}} I^{(\frac{\alpha}{\beta} + \frac{\alpha}{\beta})/2 - 1} K_{\frac{\alpha}{\beta}}(2\sqrt{\alpha\beta}I); \quad (1)$$

[0086] In the above expression, $K_{\text{sub}.n}(\text{Math.})$ is used to denote Bessel's function of n th order and $\Gamma(\text{Math.})$ is used to denote the Gamma function whose dependence upon the propagation distance z is expressed as by the following equation:

$$[00002] \Gamma(z) = \int_0^\infty \exp(-t)t^{z-1} dt; \quad (2)$$

[0087] If it is considered that the wave propagating through FSO link as a plane wave, the intensity fluctuations α and β may be expressed in terms of the link parameters as given by:

$$[00003] \alpha = [\exp(\frac{0.49 \frac{\sigma^2}{I}}{(1 + 1.11 \frac{\sigma^2}{I})^{1.17}}) - 1]^{-1}; \quad (3) \quad \beta = [\exp(\frac{0.51 \frac{\sigma^2}{I}}{(1 + 0.69 \frac{\sigma^2}{I})^{0.833}}) - 1]^{-1}; \quad (4)$$

[0088] In the expressions above, $\sigma_{\text{sub}.l.\text{sup}.2}$ represents intensity variance and is directly dependent upon the magnitude of atmospheric turbulence. The intensity variance depends upon the FSO link parameters such that $\sigma_{\text{sub}.l.\text{sup}.2} = 1.23 C_{\text{sub}.n.\text{sup}.2} k_{\text{sup}.7/6} L_{\text{sup}.11/6}$, where L is the range of FSO link and $k = 2\pi/\lambda$ represents the wave number. The parameter $C_{\text{sub}.n.\text{sup}.2}$ is known as the refractive index structure parameter and accounts for the variations in the refractive index of the link due to different weather conditions. For weak turbulence, the value of $C_{\text{sub}.n.\text{sup}.2}$ is around $10_{\text{sup}.-17}$, while for strong turbulence, its value can be as high as $10_{\text{sup}.-12}$. The performance of the disclosed FSO link under different turbulence conditions is described below.

[0089] As described earlier, an optical DQPSK signal is generated at MDC **102** by differentially encoding two separate data streams where each stream has a bit rate of 20 Gbps. Since phase modulation does not affect the amplitude of the optical pulses, the third channel **222** may be transmitted by modulating the amplitude of the resulting DQPSK optical signal. Therefore, the DQPSK signal is further amplitude modulated by a third 20 Gbps data stream to implement PAM-2 modulation. PAM-2 modulation assigns different amplitudes to the '0' and '1' bits in the data stream. Therefore, the phase information stored in each optical pulse remains intact. The EAM **224** is used for implementing PAM-2 due to cost efficiency and ease of integration with other semiconductor devices. In this way, a single pulsed optical signal may be used to carry combined data at the rate of 60 Gbps for three different channels.

[0090] The DQPSK plus PAM-2 modulated signal is sent over a 500 m FSO channel toward MDC-2. To represent medium turbulence conditions in a data center scenario, the refractive index structure parameter chosen for the FSO link is $C_{\text{sub}.2.\text{sup}.n} = 5 \times 10_{\text{sup}.-15} \text{ m}_{\text{sup}.-2/8}$. FIG. 5A-FIG. 5I show eye diagrams at different locations of the bidirectional link of FIG. 2. In examples, the eye diagrams (also referred to as eye pattern) is a graphical display (for example, on an oscilloscope display) of a serial data signal with respect to time that shows a pattern that resembles an eye. In an example, a digital signal from a receiver is repetitively sampled and applied to a vertical input (y-axis), while a data rate is used to trigger the horizontal sweep (x-axis) leading to a pattern that many times appears like an eye. The eye diagrams are known to be an effective tool for an evaluation of the combined effects of channel noise, dispersion, and inter-symbol interference on the performance of a baseband pulse-transmission system.

[0091] The eye diagram of the received optical signal at MDC **104** after transmission through the 500 m FSO link is shown in FIG. 5A. It may be observed from the eye diagram that the FSO link has induced some intensity fluctuations over the optical signal due to atmospheric turbulence. Even in the presence of intensity fluctuations, the two intensity levels representing PAM-2 modulation are still separable. Therefore, a part of this received optical signal at MDC **104** may be used to extract the data for the third channel **222** that was transmitted through PAM-2 modulation from

MDC **102**. FIG. **5B** and FIG. **5C** illustrate eye diagrams of the in-phase and quadrature components of the received optical signal after demodulation. It may be observed that due to balanced detection at MDC **102**, the eye diagrams of FIG. **5B** and FIG. **5C** are obtained as the sum of two opposite polarity Gaussian-shaped pulses. Therefore, the eye-opening is wide even though two different intensity levels are seen due to the implementation of PAM-2 over the optical signal.

[0092] As described, a part of the received optical signal is passed through the SPM-based wavelength converter **261** for suppression of intensity fluctuations as well as the generation of three separate optical signals for the transmission of data from MDC-**104** to MDC-**102**. FIG. **5D**, FIG. **5E**, and FIG. **5F** show eye diagrams of the optical signals obtained at the output of the off-set filters centered at wavelengths of 1550.6 nm, 1552.5 nm, and 1553.5 nm, respectively. It is evident from the eye diagrams that the intensity variations over the pulses have decreased to a greater extent after passing through the wavelength converter. The wavelength converter **261** also reshapes the signals by filtering a broadened spectrum that is available at the output of the HNLF **266**. The three signals are intensity modulated separately by three different uplink channels through MZMs **272.sub.1-272.sub.3** to generate OOK signals while each signal has a bit rate of 20 Gbps. The OOK signals at different wavelengths are combined and transmitted toward MDC **104** where direct detection is performed. The eye diagrams of the direct detected uplink signals for the first channel **204**, the second channel **206**, and the third channel **222** are shown in FIG. **5G**, FIG. **5H**, and FIG. **5I**, respectively. The eye-opening is quite wide for the three signals since the distance covered through free space is only 500 m and the turbulence considered is also medium.

[0093] FIG. **6** shows the BER versus received optical power plots for the downlink (the MDC **102** to the MDC **104**) and uplink (the MDC **104** to the MDC **102**) signals under medium turbulence conditions. It can be noticed from the BER curves that uplink signals (plot line **608**, plot line **610** and plot line **612**) have better receiver sensitivities compared to the downlink signals (plot line **602**, plot line **604** and plot line **606**). The reason may be due to the type of modulation used in the two directions. The use of PAM-2 divides the intensity of the downlink optical pulses into two levels, resulting in reduced eye-opening for PAM-2 as well as for the DQPSK signals. The BER of the PAM-2 signal may be improved by increasing the difference between the two intensity levels representing ones and zeros. However, this will result in an increase in the BER of the in-phase and quadrature components of the DQPSK signals since their eye-opening will be reduced consequently. Therefore, an optimum difference between the two intensity levels of the PAM-2 signal is chosen such that the BERs of all three uplink channels are close to each other.

[0094] FIG. **6** shows that for the downlink channels, a BER of 10^{-9} is achieved when the received optical power is around -13 dBm. On the other hand, the uplink channels have a BER of 10^{-9} when the received optical power is around -23 dBm. This results in a power penalty having a value of around 10 dBm between the uplink and downlink channels. As described, the uplink channels are transported by using optical pulsed signals whose intensity fluctuations are suppressed by the wavelength converter. Each channel is used to intensity modulate a single pulsed optical signal, resulting in wide eye openings after direct detection at MDC-2.

[0095] The BER results in FIG. **6** show that the performance of the disclosed FSO link is acceptable even with the use of a single laser source for the bidirectional transmission of three different channels. The performance of the link of the disclosure is compared with a conventional setup where each channel is transmitted using a separate pulsed optical source. The separate pulsed optical sources are centered at wavelengths of 1551 nm, 1552 nm, and 1553 nm, respectively and are intensity modulated by three separate data sources, each having a data rate of 20 Gbps. The resulting BER plots are also shown in FIG. **6** for comparison. The results show that the performance of the conventional FSO link using separate pulsed optical sources (plot line **612**, plot line **614** and plot line **616**) for the transmission of data under similar optical link parameters is very close to the performance of the link of the disclosure that uses regenerated pulsed optical signals. This implies that the regenerative wavelength converter reduces the intensity fluctuations

effectively. In the embodiments herein, practical values for the parameters of all the components are used for the bi-directional optical communication system **200** including the FSO link. As seen from the figures, the bi-directional optical communication system **200** of the disclosure provides a high data rate and cost-efficient solution for network designers, especially at data centers. In the disclosure, the bi-directional optical communication system **200** has been described for MDC scenario to demonstrate the working principle. While an MDC scenario is suitable for implementation due to its mobility and shorter installation time, the disclosed bi-directional optical communication system **200** may be implemented for any high data rate.

[0096] An exemplary 60 Gbps bidirectional FSO link, as disclosed herein, is suitable for a data center scenario where the data rates are generally high, and the atmospheric turbulence is present. A single-mode high repetition rate pulsed optical signal is an expensive resource; therefore, it is desirable to employ the minimum number of these sources to achieve cost efficiency while maintaining high data rates. This is achieved by the system of the disclosure by first modulating the pulsed optical source using DQPSK modulation and then demodulating it to achieve PAM-2 signal. In this way, large data can be transmitted from one MDC to another in the downlink direction. Furthermore, the same optical carrier is reused by passing it through a regenerative wavelength converter to generate three pulsed optical carriers at different wavelengths. These carriers are employed to send uplink data at the same rate of 60 Gbps as the downlink data. The BER results show that the high data rate FSO link of the disclosure is a substitute for fiber-based optical interconnects in data centers or point-to-point links in PONs, enabling high data rate bidirectional transmission.

[0097] Numerous modifications and variations of the present disclosure are possible in light of the above teachings. It is therefore to be understood that within the scope of the appended claims, the invention may be practiced otherwise than as specifically described herein.

Claims

1. An optical communication system for passive optical networks with high data rate communication, comprising: a first optical data processing unit, the first optical data processing unit comprising: a first transmitter, the first transmitter comprising: a first channel configured to provide a first electrical signal at a first data rate; a second channel configured to provide a second electrical signal at the first data rate; wherein the second data rate is two times the first data rate; a differential quadrature phase shift keying (DQPSK) encoder configured to encode the first electrical signal and the second electrical signal to provide at least two differentially encoded electrical signals at the first data rate; a Gaussian pulse generator configured to generate an optical pulsed signal centered at a wavelength with a pulse duration; an optical power splitter configured to split the optical pulsed signal into at least two optical pulsed signals; at least two Mach-Zehnder modulators (MZM) configured to phase modulate the at least two optical pulsed signals using the at least two differentially encoded electrical signals to provide a first phase modulated optical pulsed signal and a second phase modulated optical pulsed signal; a phase shifter configured to phase shift the first phase modulated optical pulsed signal to generate a phase shifted phase modulated optical pulsed signal; an optical combiner configured to combine the second phase modulated optical pulsed signal and the phase shifted phase modulated optical pulsed signal to generate a processed optical pulsed signal at a second data rate; a third channel configured to provide a third electrical signal at the first data rate; an Electro-absorption Modulator (EAM) configured to amplitude modulate the processed optical pulsed signal using the third electrical signal to provide a phase-amplitude modulated optical pulsed signal at a third data rate, wherein the third data rate is three times the first data rate; an optical amplifier configured to amplify the pulse-amplitude modulated optical pulsed signal and generate a transmitter optical pulsed signal for transmission; and wherein the transmitter optical pulsed signal is transmitted as a first free space optical link through a first

transmitter telescope; and a second optical data processing unit, the second optical data processing unit comprising: a first receiver, the first receiver comprising: a first receiver telescope configured to receive an optical pulsed signal, wherein the receiver optical pulsed signal is an attenuated version of the transmitter optical pulsed signal; a first optical splitter configured split the optical pulsed signal into a receiver optical pulsed signal and a reuse optical pulsed signal; a second optical splitter configured to split the receiver optical pulsed signal into a first optical pulsed signal, a second optical pulsed signal and a third optical pulsed signal; a photodetector configured to convert the first optical pulsed signal into a first electrical signal with two amplitude levels corresponding the third channel in the first transmitter; a differential quadrature phase shift keying (DQPSK) demodulator configured to demodulate the second optical pulsed signal and the third optical pulsed signal into a first amplitude-modulated electrical signal and a second amplitude-modulated electrical signal; at least three low-pass filters configured to remove harmonics from the first electrical signal, the first amplitude-modulated electrical signal and the second amplitude-modulated electrical signal to provide three filtered electrical signals; and at least three bit-error rate analyzers configured to estimate a bit error rate of each of the three filtered electrical signals.

2. The optical communication system of claim 1, wherein the second optical data processing unit further comprises: a wavelength converter configured to utilize the reuse optical pulsed signal for further transmission, comprising: a first optical amplifier configured to receive the reuse optical pulsed signal and further configured to suppress amplitude variations of the reuse optical pulsed signal; an optical bandpass filter tuned at the wavelength configured to remove out-of-band noise and generate a filtered optical pulsed signal; and a highly nonlinear fiber (HNLF) configured to self-phase modulation-induced spectral broadening of the filtered optical pulsed signal to generate a converted optical pulsed signal; and a second transmitter configured, the second transmitter comprising: an optical splitter configured to split the converted optical pulsed signal into three optical pulsed signals; at least three optical bandpass filters configured to filter the three optical pulsed signal into three filtered optical pulsed signals, wherein each of the at least three optical bandpass filters are at an off set from the wavelength; at least three channels wherein each channel provides an electrical signal at a first data rate; at least three Mach-Zehnder modulators (MZM) configured to On-Off keying (OOK) modulate the three filtered optical pulsed signals using the three electrical signals to provide three modulated optical pulsed signals; an optical combiner configured to combine the three modulated optical pulsed signals into a modulated optical pulsed signal; a second optical amplifier configured to amplify the modulated optical pulsed signal and generate a second transmitter optical pulsed signal; and wherein the second transmitter optical pulsed signal is transmitted as a second free space optical link through a second transmitter telescope.

3. The optical communication system of claim 1, wherein the first optical data processing unit further comprises: a second receiver, wherein the second receiver comprises: a second receiver telescope configured to receive an optical pulsed signal; an optical splitter configured to split the received optical pulsed signal into three optical pulsed signals; at least three optical bandpass filters centered at the off set from the wavelength configured to filter the three optical pulsed signals to provide three filtered optical pulsed signals; at least three photodetectors configured to convert the three filtered optical pulsed signals into three electrical signals; at least three low-pass filters configured to remove harmonics of the three electrical signals; and at least three bit error rate analyzers configured to estimate a bit error rates of each of the three electrical signals.

4. (canceled)

5. (canceled)

6. The optical communication system of claim 1, wherein the wavelength is 1552 nm.

7. The optical communication system of claim 1, wherein the electro-absorption modulator (EAM) utilizes a non-return-to-zero (NRZ) pulsed signal at the first data rate to implement pulse amplitude modulation.

- 8.** The optical communication system of claim 1, wherein the phase-amplitude modulated optical pulsed signal is encoded as a differential quadrature phase shift keying (DQPSK) and two-level pulse amplitude modulation (PAM-2).
 - 9.** The optical communication system of claim 1, wherein the differential quadrature phase shift keying (DQPSK) demodulator further comprises: a plurality of delay lines; a plurality of phase shifters, wherein the plurality of delay lines and the plurality of phase shifters are configured to compare a bit of a received optical pulsed signal to a previous bit of the received optical pulsed signal; and a group of four detectors configured to employ balanced detection of the received optical pulsed signal.
 - 10.** The optical communication system of claim 1, wherein the three filtered electrical signals in the first receiver correspond to a data of the first channel, the second channel and the third channel in the first transmitter.
 - 11.** The optical communication system of claim 2, wherein the wavelength converter is a self-phase modulation (SPM)-based wavelength converter.
 - 12.** The optical communication system of claim 2, wherein the wavelength converter is configured to reuse a received optical signal for further transmission of electrical signals.
 - 13.** The optical communication system of claim 2, wherein the off set of the wavelength are three wavelength values including 1550.6 nm, 1552.5 nm, and 1553.3 nm.
 - 14.** The optical communication system of claim 3, wherein the three electrical signals at the second receiver correspond to the electrical signals of the three channels at the second transmitter.
 - 15.** The optical communication system of claim 1, wherein the optical bandpass filter of the wavelength converter is configured to remove noise in the form of an amplified spontaneous emission (ASE) noise.
 - 16.** The optical communication system of claim 1, wherein the first optical data processing unit and the second optical data processing unit are each at least one selected from a data center and a modular data center.
 - 17.** The optical communication system of claim 1, wherein the first free space optical link is a point-to-point free space optical link.
 - 18.** The optical communication system of claim 1, wherein the first free space optical link is at a data rate of three times the first data rate.
 - 19.** The optical communication system of claim 1, wherein the first free space optical link is at least one from a unidirectional free space link and a bidirectional free space link.
 - 20.** The optical communication system of claim 2, wherein the second free space optical link is a modified version of the first free space optical link.
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